MGMT-38900
Using AI for Business Analytics

PURDUE UNIVERSITY | Mitchell E. Daniels, Jr. School of Business

COURSE NUMBER: MGMT 83900 – Using AI For Business Analytics
(CRN 21501)

Semester Year*:** Spring 2026 ***Credit Hours: 3
Meeting Time*:** TR 10:30-11:45 am. ***Location: **RAWL 4082**
Prerequisites: MGMT 305 Business Analytics or ECON 260 or equivalent statistics course.
Brightspace Course Link: https://purdue.brightspace.com/d2l/home/1497261

CONTACT INFORMATION & OFFICE HOURS

Instructor: Juan Ignacio Gonzalez, Ph.D. **Office:** **KRAN 533**
Office hours: Wednesdays, 10:00 to 11:30 AM (in person).
Contact: gonz1400@purdue.edu
Expected email response time: Email responses are typically provided within 24–48 hours (often sooner). If you do not receive a reply within 48 hours, please follow up. I generally do not respond to emails after 9 pm. Please make sure you contact me earlier than 9 pm on the night before assignments are due to ensure you get a response.
NOTE: Assistance by email or appointment ***is also available***. Contact me by email to arrange an appointment. Please avoid "last-minute" contact before assignments are due to ensure you receive a timely response.

COURSE DESCRIPTION

Course Objective:

In this course, you will ***explore and learn how artificial intelligence empowers the development of data-driven solutions supporting business decision-making***. ***You will integrate AI tools, programming environments, and analytical techniques to design and deploy intelligent solutions for real business challenges***. Emphasizing experimentation, teamwork, and creative problem-solving, ***the course equips students with the technical and foundational analytical skills needed to transform data into actionable insights*** and to thrive in AI-enhanced decision environments.

Learning Outcomes:

Upon completing this course, students will be able to:

- ✓ ***Apply diverse analytics platforms and programming languages*** to process, visualize, and interpret business data.
- ✓ ***Use artificial intelligence tools***, including generative AI and low/no-code environments, ***to create data-driven business solutions.***

- ✓ ***Collaborate to solve business challenges with analytical rigor, creativity, and strategic insight***.
- ✓ ***Communicate analytical results through clear, professional reports and presentations for academic and business audiences.***
- ✓ Participate in conferences, academic forums, and professional events to showcase the tangible outcomes from the course.
- ✓ ***Design and maintain an organized professional portfolio*** website showcasing projects, code, and posters on GitHub.

Course Structure and modality:
This course is delivered in an in-person, face-to-face format and is designed to promote active learning through a combination of instructor-led discussions, hands-on analytical exercises, collaborative team activities, and applied projects. Class sessions will integrate short conceptual explanations with practical work using data, analytical tools, and GenAI applications, allowing students to apply concepts immediately. Students are expected to actively participate in discussions, engage with peers during team-based activities, and contribute to problem-solving exercises centered on real business scenarios. ***The course structure emphasizes experiential learning, critical thinking, and iterative feedback***, ensuring that students progressively build analytical competence and confidence in applying data-driven approaches to managerial decision-making.

COURSE RESOURCES/REQUIREMENTS

- *Learning Management System (LMS): Brightspace* – This system serves as the central hub for accessing, sharing, tracking, and archiving course-related information. It includes course notes, assignments, exams, discussions, earned scores, and more. You can log in to your Brightspace account at http://purdue.brightspace.com.
- ***Optional*** *Textbooks:*
  - ✓ *R for Everyone: Advanced Analytics and Graphics (2nd Edition)*, by Lander, et al. (ISBN-13:978-0134546926). Available through the Purdue Library.
  - ✓ Python for Data Analysis: Data Wrangling with pandas, NumPy, and IPython by Wes McKinney (ISBN-13: 978-1491957660). Available through the Purdue Library. Also available at: https://wesmckinney.com/book/
  - ✓ An additional set of bibliographical resources is available for the class. See them listed in the **Learning sources and texts** module in Brightspace.
- *Software Tools*: This course utilizes a set of widely used analytics and AI-enabled tools to support hands-on, in-class learning. Core software includes:
  - ✓ RStudio (personal desktop download) or Purdue's cloud-based Scholar-Gateway
  - ✓ Python (Jupyter notebook, personal desktop download) or Purdue's cloud-based Scholar-Gateway
  - ✓ Google Colab
  - ✓ OpenAI, Copilot, Gemini, Claude, Purdue GenAI Studio.
- *Computing (Required):* A laptop/tablet (***required*** for work in class) and a desktop (optional for

completing home assignments), with internet access and the capability to run Python and R Programs. Ensure your computer meets the following minimum requirements: **Processor (CPU):** Intel i5 (8th gen or newer) / AMD Ryzen 5 (2000 series or newer). **RAM:** 8 GB. **Operating System:** Windows 10/11, macOS 12+, or Linux Ubuntu 20+)

- **Students must bring a laptop or tablet to class to complete assigned tasks.**

ASSESSMENTS/ASSIGNMENTS & GRADING SCALE

Course Assignments: (refer to Brightspace for detailed assignments and due dates).

Beyond comprehension of theoretical concepts, this course is structured around the **Applied AI Analytics Learning Framework (A³LF)**, which emphasizes the development and assessment of students' proficiency across three integrated dimensions: **A**nalytical foundations, **AI**-enabled application, and **A**ctionable presentation and communication. In addition to technical accuracy, the course places a strong emphasis on analytical communication, professional collaboration, and effective teamwork. To excel in the course, students must not only provide accurate and well-reasoned responses in assignments and challenges but also actively engage in class discussions and team-based project activities. Overall performance is evaluated holistically using a performance-based grading structure aligned with the A³LF framework, as detailed in the Grading Scale below.

Grading Scale.

Attendance/Participation (ClassPoint activities, In-Class Submissions)	10%
Individual Case Solution	15%
Team Challenges (4)	35%
Final project (2 deliverables, mid-term: 10%, final: 25%)	35%
Completed professional profile and portfolio on GitHub	3%
Discussion forum participation and course agreement	2%
	100%

Here is a brief description of each activity to be evaluated during the course:

Attendance and Participation (10%)
Active attendance is required, as class sessions include **ClassPoint** activities, in-class submissions, guided exercises, and collaborative problem-solving. Participation and submission of correct responses (along with other in-class activities) will be used as evidence of attendance. Any problem with the **ClassPoint** platform must be reported to the instructor immediately (as soon as it happens) so it can be documented and addressed. Only students who are physically present in class will be awarded attendance and participation points.

Individual Case Solution (15%)
Each student will independently design, develop, and present a complete predictive or clustering solution addressing a specific business problem of their choice. This includes identifying and preparing an appropriate dataset, developing the analytical model, and communicating results through a final individual presentation to the instructor and a guest evaluator, either live or via recorded video. Detailed instructions for scheduling the in-person presentations (if applicable) and rubrics will be provided in Brightspace. More details on this deliverable will be available in Brightspace.

Team Challenges (4 challenges – 35%)
Team challenges are module-based applied activities aligned with the course schedule. For each challenge, teams will submit a complete analytical solution developed during the corresponding module (4 submissions at 5% each, totaling 20%). In addition, a selection of teams will present their solutions during class (2 presentations at 7.5% each, totaling 15%). Presentation teams will be announced in advance, and the evaluation will focus on analytical rigor, clarity of insights, and professional communication.

Final Project (35%)
The final project is a team-based, end-to-end applied analytics project developed in collaboration with an industrial partner. The project consists of two graded deliverables: a mid-term milestone (10%) and a final solution and presentation (25%), evaluated by the instructor and an external panel. Emphasis is placed on solution design, responsible AI use, deployment readiness, and the ability to translate analytics into actionable business insights. More details on this deliverable will be available in Brightspace.

Professional GitHub Profile and Portfolio (3%)
Each student is required to complete and submit a personal GitHub profile and professional portfolio showcasing their individual and team work from the course. This component assesses organization, completeness, and professional presentation of analytical artifacts.

Discussion Forum Participation and Course Agreement (2%)
Students are required to participate in designated discussion activities to present evidence, reflect on course topics, and contribute to the learning community. Acceptance of the course agreement is mandatory and must be completed by the specified deadline.

Grading Policy.
Final course grades are calculated on a 100-point scale and converted to official letter grades using the numeric ranges shown in the grading table below. This scale defines the equivalence between numeric performance and the letter grade reported to the Office of the Registrar and is applied consistently to all students:

A+	>=100	**B+**	87-89	**C+**	77-79	**D+**	67-69	**F**	<60

A	96-99	**B**	84-86	**C**	74-76	**D**	64-66
A-	90-95	**B-**	80-83	**C-**	70-73	**D-**	60-63

Please note: Final grades will be determined based on the relative ranking of total earned points, not solely on point counts or percentages. Final grades are curved in accordance with the Daniels School of Business, Undergraduate Grading Policy, which targets an average GPA of about 3.3 for elective courses:

Grade	***A/A-***	***B+/B***	***B-***	***C+***	***C or below***
GPA	*4/3.7*	*3.3/3*	*2.7*	*2.4*	*<2*
% of Students	*30-40%*	*30-40%*	*10-20%*	*10-15%*	*< 10%*

OTHER GRADING POLICIES

Grade review request:

Grades and assignment solutions/feedback will be released within seven calendar days after the deadline. Students who wish to request a review of a grade must do so within 7 calendar days of the grade's release.

Grade review requests must be based on legitimate concerns, such as:

- ✓ Misapplication of analytics principles,
- ✓ Errors in the logic of thought,
- ✓ Mistakes in grading accuracy.

Requests based on post-hoc legalistic arguments or general dissatisfaction with the grade will not be considered.

How to Submit a Grade Review Request:

- ✓ Review the provided solutions or feedback carefully
- ✓ If you still believe an error exists, email the instructor with the following information:
  - Course name, section and PUID
  - Lecture date/time
  - Assignment number/title
  - Specific grade or rubric item in question
  - A clear explanation of why the deduction is believed to be incorrect, referencing the solutions or grading rubric.

Note that grades will not be discussed in the classroom, before, during, or after class. Students are encouraged to attend office hours for questions related to course content or assignment clarification. After the 7-day review period (or 3-day period for specified exceptions), all grades are final and cannot be revised further for final course calculations.

Extra credit opportunities.

- **Course Evaluations.** Students will have the opportunity to evaluate both the course and the **instructor anonymously. Near the evaluation period, you will receive an official email from** Purdue's evaluation administrators containing a secure link to the online course evaluation system.
  Your feedback is highly valued and plays an important role in enhancing the quality of instruction and improving educational outcomes at Purdue University. To encourage participation, an incentive is provided:
  - ✓ *If at least 85% of the class submits the midterm and/or final course evaluations, each participating student will receive a 1% bonus, which will be added to their final Attendance and Participation score.*
    Purdue's evaluation platform tracks overall class submission rates during the official evaluation period. Although I will not receive individual responses, I will provide periodic updates on class participation rates.

- **Professional Certifications.** Students may earn extra credit by completing a professional certification related to tools used in this course (e.g., R or Python, Tableau, Power BI) during the semester. The certification must be approved by the instructor in advance by email; unapproved certifications will not be considered. Upon submission of a valid certificate issued to the student, up to 5 bonus points (out of 100) may be added to the final course grade. See more details in the **Professional Certifications** module in Brightspace. **Certifications need to be completed and granted during the semester. No certification will be considered if completed before the first day of class of Spring 2026.**

HOW TO BE SUCCESSFUL IN THIS COURSE

Engage actively in every session.
Class time is essential for participating in **ClassPoint** activities, guided exercises, and teamwork. *Example: submitting in-class responses and contributing to model-building discussions during challenge weeks directly supports attendance and participation credit.*

Prepare consistently for applied work.
Reviewing assigned materials before class allows you to fully engage in modeling, dashboards, and agent construction activities. *Example: reviewing prompt-engineering videos before class enables you to contribute effectively during in-class challenge development.*

Practice end-to-end Analytics independently.
Success requires hands-on practice beyond class, including data preparation, modeling, and interpretation. *Example: experimenting with alternative datasets or model parameters strengthens your individual case solution and final project.*

Collaborate professionally within your team.
Team challenges and the final project depend on clear communication, accountability, and shared problem-solving. *Example: Dividing modeling, validation, and presentation tasks early improves submission quality, presentation delivery, and team communication.*

Communicate early and clearly
Proactive communication helps resolve technical, analytical, or logistical issues before they affect performance. *Example: reporting ClassPoint issues immediately or asking questions during office hours prevents loss of participation credit and project delays.*

COURSE POLICIES

- All assignments are expected to be submitted on time through the designated platform and in the required format. Late submissions will be considered only when a formal extension has been approved in advance under exceptional circumstances. Requests for make-up work or extensions are reviewed on a case-by-case basis and must be based on ***reasons that would reasonably be accepted in a professional work environment***, such as documented medical situations requiring rest or recovery, family emergencies, or other unforeseen circumstances beyond the student's control. Situations including oversleeping, travel plans, personal commitments, or time management challenges generally do not qualify for make-up consideration. See here the Class Absence Policy for details.
- Participate actively in all course-related activities, including lectures, discussions, and team projects. Regularly monitor your email and Brightspace announcements to stay informed.
- Provide thoughtful and constructive feedback in all team-based assignments and peer review activities.
- Please review and acknowledge the policies, statements, and procedures relevant to our educational environment, available in the **"Purdue Policies and Statements"** section on Brightspace. Since this is not an exhaustive list, consult your academic advisor for complete and detailed information.
- Purdue University is committed to providing accessible learning experiences for all students. If you anticipate or encounter physical or academic barriers due to a disability, please inform the instructor to explore appropriate accommodations. You are also encouraged to contact the Disability Resource Center at drc@purdue.edu or 765-494-1247 for additional support.
- Read the document "Classroom Expectations at the Daniels School of Business (DSB)," available in Brightspace, to prepare for active engagement.
- **Peer evaluations will be conducted for team-based activities to assess individual contributions, professionalism, and engagement. These evaluations will be reviewed by the instructor and may be used to adjust individual grades on team assignments and the final project to reflect each student's level of contribution.**
- **Contact the instructor promptly with any questions, suggestions, or requests for assignment extensions due to justified and exceptional causes/absences.**

- **AI Policy:** You are encouraged to use AI tools that enhance your individual or group learning. Developing the ability to work effectively with AI is a valuable and emerging skill, and I am available to guide during sessions, office hours, or by appointment. Please follow these guidelines:
  - Low-effort prompts typically yield poor-quality results.
  - Use AI as a support for your learning. Request explanations, examples, or clarification of

doubts rather than direct solutions.
- Do not accept AI outputs uncritically. You are responsible for verifying accuracy and addressing any errors or omissions. ***Copy-and-paste practices are prohibited.***
  - **Always acknowledge AI use.** Include a brief note at the end of any assignment explaining how AI supported your work and properly cite any AI-generated sources.

➢ Any incorrect use will be considered an issue of Academic Integrity.

ACADEMIC DISHONESTY

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Part 5, Section III-B-2-a, University Regulations]. Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972]

Purdue's Honor Pledge: "As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue." **Please adhere to these definitions and recommendations.**

Lastly, do not post notes, assignments, codes, handouts, or exams on coursehero.com, koofers.com, or other "study" websites. ***You do NOT have my permission****. These materials are considered derivative works of the instructor and are therefore subject to the instructor's copyright.*

SEMESTER OUTLINE

The class schedule is outlined below. Specific instructions, due dates and times, and rubrics for each activity will be available in Brightspace.

MODULE	WEEK	DAYS	TOPIC	READINGS /VIDEOS	ASSIGNMENTS
Module1. Welcome and set up	**W1**	**13-Jan**	**1.1 Course Introduction. Class dynamics and syllabus overview.**	**-**	- Set up accounts for GenAI interfaces. - Configure your GitHub account.
		15-Jan	**1.2 Why studying these topics matters. GenAI vs. ML vs. DL: key differences I.**	**-**	- Sign the course agreement. (All due 1/18)
Module 2. Foundations of GenAI and LLMs	**W2**	**20-Jan**	**2.1 GenAI vs. ML vs. DL: key differences II.** **2.2 What is an LLM?**	**Set of videos and readings:**	**-**
		22-Jan	**2.3 Neural networks and the Transformer architecture.**	**week 2. Due on 1/19**	**-**

MODULE	WEEK	DAYS	TOPIC	READINGS /VIDEOS	ASSIGNMENTS
			2.4 Text vectorization and embeddings. **2.5 An integrated view of an LLM (simulation)**		
Module 3. **Prompt Engineering and Algorithmic Thinking**	**W3**	**27-Jan**	**3.1 Communicating effectively with LLMs.**	**Set of videos**	**-**
		29-Jan	**3.2 Prompt Engineering.**	**and readings: weeks 3 & 4.**	**-**
	W4	**3-Feb**	**3.3 Algorithmic Thinking.**	**-**	**-**
		5-Feb	**3.4 Prompting for data analysis and predictive analytics.**	**-**	**-**
Module 4. Exploration and description	**W5**	**10-Feb**	**4.1 Data exploration and wrangling**	**Set of videos and readings:**	**-**
		12-Feb	**4.2 Data visualization** **4.3 Dashboards and descriptive analytics**	**week 5. Due on 2/9**	-
	W6	**17-Feb**	**4.4 Summary: Data exploration and correction. Boston Housing exercise**	**-**	-
		19-Feb	**Keynote Speaker: Noam Eppen**		**-**
	W7	**24-Feb**	**4.5 Dashboards and data description.** 4.5.1 Goal definition 4.5.2 Prompt design 4.5.3 Publication	**Set of videos and readings:**	**-**
		26-Feb	**Helping Thursday:** 1. Work on Challenge 1 2. Orientation on midterm submissions for Team and individual semester projects.	**week 7.**	-
	W8	**3-Mar**	**CHALLENGE 1: Interactive dashboard presentations.**	**Set of videos and readings:**	All teams submit dashboards (.ipynb / R Script /.pbix) by the end of the day on 3/2 . Selected teams present their dashboards during class on 3/3.

MODULE	WEEK	DAYS	TOPIC	READINGS /VIDEOS	ASSIGNMENTS
Module 5. Prediction (numeric and categorical variables)		5-Mar	**5.1 Predicting numerical outcomes** 5.1.1 Prediction, performance, and error 5.1.2 Models for predicting numerical outcomes 5.1.3 Fine-tuning and model selection	week 8	-
	W9	10-Mar	**5.1 Predicting categorical outcomes** 5.1.1 Prediction, performance, and error 5.1.2 Models for predicting numerical outcomes 5.1.3 Fine-tuning and model selection	Set of videos and readings:	-
		12-Mar	**Helping Thursday:** 1. Work on Challenge 2 2. Orientation on final submissions for team and individual semester projects.	week 9	Deadline for mid-term presentations of team AND individual projects: 3/13
	W10	17-Mar 19-Mar	**SPRING BREAK. No classes.**		
Module 6. Unsupervised modeling: clustering	W11	24-Mar	**CHALLENGE 2: Predictive modeling presentations.**	Set of videos and readings:	**All teams submit models** (.ipynb / R Script) by the end of the day on 3/23. **Selected teams present** their predictive models during class on 3/24.
		26-Mar	**6.1 Clustering: goals and business benefits** **6.2 Distance measures** **6.3 Profiling and description**	week 11	-
	W12	31-Mar	**Helping Tuesday:** 1. Work on Challenge 3 2. Orientation on final submissions for team and individual semester projects.	-	-
		2-Apr	**CHALLENGE 3: Clustering model presentations.**	-	**All teams submit models** (.ipynb / R Script) by the end of the day on 4/1. **Selected teams present** their clustering models during class on 4/2.
Module 7. AI Agents.	W13	7-Apr	**7.1 AI and the agentic perspective**		

MODULE	WEEK	DAYS	TOPIC	READINGS /VIDEOS	ASSIGNMENTS
			7.2 Agent construction **7.3 Multi-agent systems**	**Set of videos and readings:**	
		9-Apr	**Helping Thursday:** 1. Work on Challenge 4 2. Orientation on final submissions for team and individual semester projects.	**week 13**	
Module 8. Ethics and responsible AI.	**W14**	**14-Apr**	**CHALLENGE 4: Presentations of agent integrations to models.**	**Set of videos and readings:**	**All teams submit model** integrations (.ipynb / R Script) by the end of the day on 4/13. **Selected teams present** their model integrations during class on 4/14.
		16-Apr	**8.1 Bias, Fairness, and Risk in AI-Driven Decisions** **8.2 Transparency, explainability, and responsible deployment**	**week 14**	
Module 9. Final Project preparation.	**W15**	**21-Apr**	**Week 15 sessions are to be devoted to orienting teams on final project preparation:**	**-**	Check in with each team for instructor validation of final presentations
		23-Apr	9.1 Model training and fine-tuning. 9.2 Model deployment and publishing. 9.3 Final presentation preparation.	**-**	
	W16	**28-Apr**	**Final Project presentations. Part 1**	**-**	Deadline for submitting videos of individual case solutions: 5/3.
		30-Apr	**Final Project presentations. Part 2**	**-**	Teams assigned will present during the sessions.

DISCLAIMER

Activities or due dates may be adjusted if necessary. Any changes will be announced in class and updated in the Brightspace calendar.